

PhilaLink: Project Specification

"Linking Communities to Healthier Lives"

A Comprehensive Digital Health Ecosystem for South African Communities

1. Executive Summary & Vision

PhilaLink is a community-centric healthcare solution designed for the South African public health landscape. It addresses the critical challenge of chronic medication adherence and diagnostic accessibility in underserved areas. The system digitizes the manual "Proxy" collection system (CCMDD), provides automated health reminders, and leverages AI for preliminary symptom assessment and self-diagnostic guidance.

The "Mom" Story: Inspired by community health caregivers who manage medication collection for multiple elderly citizens, PhilaLink replaces physical collection cards with a digital dashboard, eliminating the risk of missed dates and nurse-patient friction.

2. Technical Architecture

Primary Technology Stack:

Layer	Technology	Purpose
Frontend	React.js	Component-based UI, state management via Context API / Redux
Backend	C# ASP.NET Core	3-Tier architecture (Data, Business, Web API); RESTful services
Database	Microsoft SQL Server	Relational storage, Geography data types, Entity Framework Core
Real-Time Comms	SignalR	Real-time updates (collection status, notifications)
AI Diagnostics Engine	Anthropic Claude Sonnet API	Powers PhilaChat's conversational self-diagnostic reasoning and OTC medication recommendations
Integration Scripting	Python	Data preprocessing and pre/post-processing glue around the Sonnet API (age-band and allergy filtering)
Geo-Intelligence API	OpenWeatherMap API	Local climate data to drive proactive, condition-specific health advice
Mapping API	Google Maps API	Clinic geolocation, distance calculation, and directions for the Clinic Finder

System Tiers:

- **Presentation Tier (React):** Responsive dashboard for Nurses, Proxies, and Patients.
- **Business Logic Tier (C#):** The "Brain" containing weighted diagnostic algorithms, background workers, and weather logic.
- **Data Access Tier (SQL Server / EF Core):** Entity Framework Core for secure and optimized database transactions.

3. Core Features & Functionality

A. Proxy Caregiver Management

A dedicated dashboard for “Nominated Proxies” (Caregivers) to manage multiple patients. It tracks collection dates for an entire group, allowing for batch collections at clinics.

B. Automated SMS Reminder Service

A C# Background Worker service that executes daily at 08:00 AM. It queries the database for all medication collections due in 48 hours and sends automated SMS notifications to both the Proxy and the Patient.

C. Weighted Symptom Diagnostic Engine

Users input symptoms (Fever, Shaking, etc.). The system calculates a weighted score against known conditions to suggest the most likely issue and provide dosage advice.

D. Geo-Environmental Intelligence

Integration with OpenWeatherMap API. The system provides proactive health advice based on local climate (e.g., advising asthma patients during high-wind days in Gqeberha).

E. Smart Clinic Finder

Uses Google Maps API to locate the nearest clinic. Features include real-time “Open/Closed” status based on server-time checks against clinic business hours.

F. PhilaChat — AI-Powered Self-Diagnostic Chatbot

Summary: PhilaChat is powered by Anthropic's Claude Sonnet API, which serves as the core inference engine for AI-driven symptom analysis and OTC medication recommendations.

PhilaChat is a conversational assistant embedded across the platform, powered by **Anthropic's Claude Sonnet API**, that guides patients through a preliminary self-diagnostic conversation. Unlike the static Symptom Diagnostic Engine (Feature C), PhilaChat conducts a natural-language dialogue, reasons over structured patient context, and returns an over-the-counter (OTC) medication recommendation where appropriate.

Key inputs and behaviour:

- **Age input:** Used to filter recommendations that are age-appropriate (e.g. excluding adult-dosed medication for pediatric or elderly users) and to flag when a case should be escalated to a Nurse instead of self-managed.
- **Allergy input:** Cross-references the patient's declared allergies (linked to the existing **Allergy** entity) against active ingredients in candidate OTC medications before the Sonnet API's recommendation is surfaced, preventing unsafe suggestions.
- **Symptom conversation:** The Claude Sonnet API reasons over the same weighted-condition data as Feature C, but delivers it conversationally rather than via a static form.
- **OTC medication recommendation:** Suggests generally available, non-prescription medication (e.g. antihistamines, analgesics, antipyretics) with dosage guidance, and always includes a disclaimer directing the user to a Nurse or clinic for anything outside OTC scope.

- **Escalation path:** Any high-risk combination of symptoms, age, or allergy conflict routes the conversation to a live Nurse referral rather than an automated recommendation.

Note: PhilaChat's OTC recommendation logic is treated as a safety-critical feature. All recommendations are logged (see Section 7, Audit Logs) and are advisory only — they do not replace clinical diagnosis.

4. Database Design (SQL Schema)

The schema below reflects the current implementation in the **PhilaLink_Backend** repository, which has expanded significantly from the original five-table concept into a full domain model covering identity, clinical, scheduling, and audit concerns.

Entity	Key Fields (representative)	Description
User	UserID, Username, PasswordHash, Role	System-wide authentication and role-based access control (Nurse / Proxy / Patient / Admin).
Admin	AdminID, UserID (FK)	Administrative account with platform-wide management privileges.
Nurse	NurseID, UserID (FK), ClinicID (FK)	Clinical staff profile; authorises collections and logs audit entries.
Proxy	ProxyID, UserID (FK)	Nominated caregiver profile responsible for one or more patients.
ProxyLink	ProxyLinkID, ProxyID (FK), PatientID (FK)	Many-to-many linking table connecting Proxies to the Patients they manage.
Patient	PatientID, FullName, IDNumber, ProxyID (FK), ClinicID (FK)	Profiles of people receiving care.
Clinic	ClinicID, Name, Latitude, Longitude, OpenTime, CloseTime	GIS data for the Smart Clinic Finder feature.
Allergy	AllergyID, PatientID (FK), AllergenName, Severity	Patient-declared allergies; referenced by PhilaChat before issuing OTC recommendations.
MedicalCondition	ConditionID, Name, Description	Master list of known medical conditions used by the diagnostic engine and PhilaChat.
SymptomMap	SymptomID, ConditionID (FK), Weight (1–10)	Core weighting table mapping symptoms to conditions for both the Diagnostic Engine and PhilaChat.
SymptomAssessment	AssessmentID, PatientID (FK), Symptoms, ResultConditionID, Timestamp	Stores the outcome of each diagnostic or PhilaChat session for clinical follow-up.
Medication	MedicationID, Name, ActiveIngredient, IsOTC, RecommendedDosage	Master medication catalogue, flagged by OTC eligibility; cross-checked against Allergy records.
Collections (MedicationSchedule)	CollectionID, PatientID (FK), ScheduledDate, Status	Tracks chronic medication collection cycles (Pending / Collected).
MedicationLog	LogID, PatientID (FK), MedicationID (FK), NurseID (FK), Timestamp	Timestamped record of every medication collection or dispensing event, tied to the authorising Nurse.
Notification	NotificationID, UserID (FK), Message, Type, SentAt, ReadStatus	Stores SMS / in-app reminder and alert records generated by the Automated Reminder Service.
OtpVerification	OtpID, UserID (FK), Code, ExpiresAt, Verified	One-time-pin records used for identity verification during registration and sensitive actions.

Entity	Key Fields (representative)	Description
AuditLog	AuditLogID, UserID (FK), Action, EntityAffected, Timestamp	System-wide accountability log capturing every sensitive action, per the POPIA compliance requirement in Section 7.

Entities such as Admin, Allergy, AuditLog, MedicalCondition, MedicationLog, MedicationSchedule, Notification, OtpVerification, ProxyLink, SymptomAssessment, and Medication have been added since the original five-table concept, reflecting the current state of the PhilaLink_Backend repository.

5. The Logic “Brain”: Weighted Algorithm

Below is a conceptual example of the C# Service that handles the diagnosis:

```
public class DiagnosisResult {
    public string ConditionName { get; set; }
    public int TotalScore { get; set; }
}

// Logic: Calculate total weight for each condition based on selected symptoms
var scores = _context.SymptomMaps
    .Where(sm => userSymptomIds.Contains(sm.SymptomId))
    .GroupBy(sm => sm.Condition.Name)
    .Select(g => new DiagnosisResult {
        ConditionName = g.Key,
        TotalScore = g.Sum(x => x.Weight)
    })
    .OrderByDescending(r => r.TotalScore);
```

PhilaChat Recommendation Logic (extension)

PhilaChat sends the SymptomMap weighting above, along with the patient's age and declared allergies, as structured context to the **Anthropic Claude Sonnet API**, which performs the conversational reasoning and produces a candidate recommendation. A Python-based post-processing step then re-validates the response against the Medication catalogue's ActiveIngredient and IsOTC fields before anything reaches the patient, ensuring only safe, non-prescription suggestions are surfaced.

6. Future Roadmap: AI Vision Assessor

Phase 2 involves integrating Azure AI Vision. This will allow users to upload images of symptoms (e.g., skin rashes) or video of physical tremors. The system will use OCR to read medication labels to ensure the correct dosage is being logged, reducing human error.

PhilaChat is positioned as the natural entry point for this roadmap: once AI Vision is integrated, image and video inputs will feed into the same conversational flow already used for age, allergy, and symptom collection.

7. Security & Compliance (POPIA)

- **Data Encryption:** All patient data is encrypted at rest using SQL TDE.
- **Privacy:** No sensitive medical images are stored on the server; analysis happens in volatile memory.
- **Audit Logs:** Every medication collection log is timestamped with the Nurse ID for accountability.
- **PhilaChat Session Logging:** Every PhilaChat conversation and OTC recommendation is persisted to SymptomAssessment and AuditLog, ensuring recommendations remain traceable and reviewable by clinical staff.